

# Accounting and Accountability

Week 9



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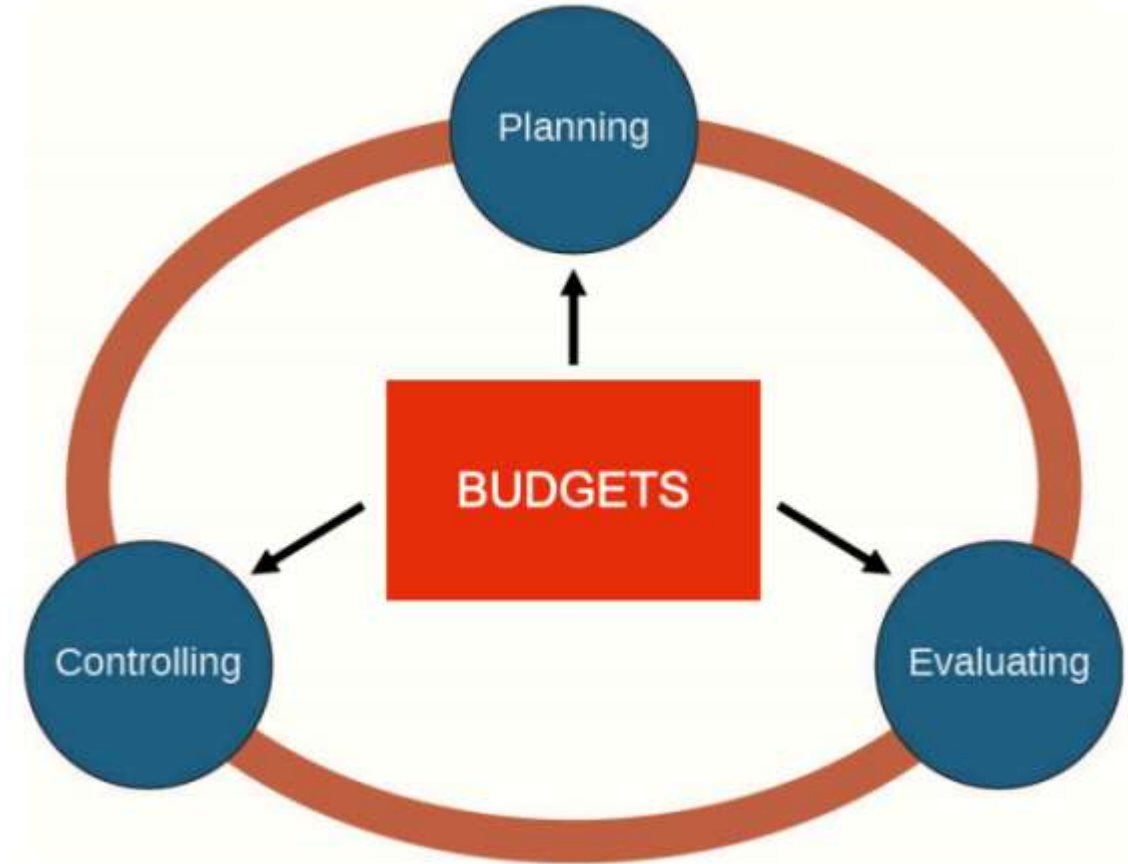
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# Key concepts

## Describe how and why managers use budgets

- The business will develop strategies and plans to help them achieve these goals. Budgeting helps businesses consider the costs of their strategies and plans (over both the short and longer term).
- A good budgeting system will help a company reach its strategic goals by allowing management to plan and to control major categories of activity, such as revenue, expenses, and predicting and managing the cash required for these activities.
- Budgets play a role in planning, controlling and evaluating.



# Key concepts

There are many **advantages** to budgeting, including:

## **Communication**

- Budgeting is a formal method to communicate a business's plans to its internal stakeholders, such as executives, department managers, and others who have an interest in—or responsibility for—monitoring the business's performance.
- Budgeting requires managers to plan for both revenues and expenses.

# Key concepts

## Planning

- Preparing a budget requires managers to consider and evaluate
  - The assumptions used to prepare the budget.
  - Long-term financial goals.
  - Short-term financial goals.
  - The business's position in the market.
  - How each department supports the strategic plan.
- Preparing a budget requires departments to work together to
  - Determine realisable sales goals.
  - Compute the manufacturing or other requirements necessary to meet the sales goals.
  - Solve bottlenecks that are predicted by the budget.
  - Allocate resources so they can be used effectively to meet the sales and manufacturing goals.
  - Compare forecasted or flexible budgets with actual results.

# Key concepts

## Evaluation

- When compared to actual results, budgets are early alerts and they forecast:
  - Cash flows for various levels of production.
  - When loans may be required or when loans may be reduced.
- Budgets show which areas, departments, units, and so forth, are profitable or meet their appropriate goals. Similarly, they also show which components are unprofitable or do not reach their anticipated goals.
- Budgets set defined benchmarks that may be used for evaluating business and management performance, including raises and bonuses, as well as negative consequences, such as firing.

# Key concepts

- There are also **disadvantages**
- Budgets can be inflexible
- They are based on assumptions which may not turn out to be correct
- They can take a long time to create – especially in large organisations where each department must prepare a budget

## Quick Check

**Which answer below is not an advantage of budgeting?**

- Budgets are a communication tool.
- Budgets help managers to plan for the future.
- Budgets help to identify issues before they happen.
- Budgets require assumptions to be made.

# Key concepts

## Understanding the structure of budgets

The **master budget** is exactly as the term describes – the budget that exists above all others. It will contain every source of revenue coming into the business and every cost the business will need to pay for.

The master budget has two major categories: the **financial budget** and the **operating budget**.

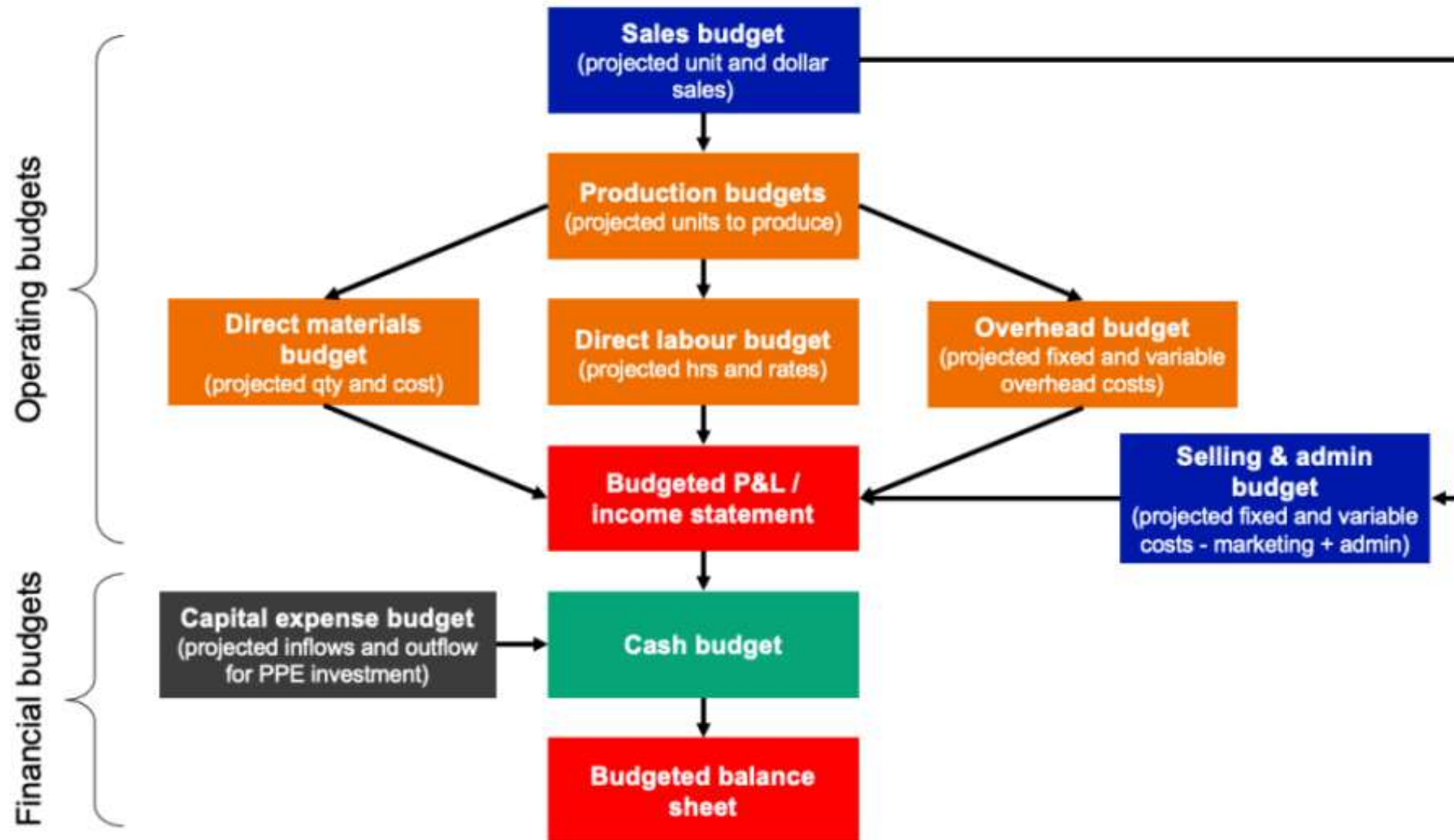
The **financial budget** plans the use of assets and liabilities and results in a projected balance sheet.

The **operating budget** helps plan future revenue and expenses and results in a projected income statement.

The operating budget has several subsidiary budgets that all begin with projected sales.

# Key concepts

## Understanding the structure of budgets



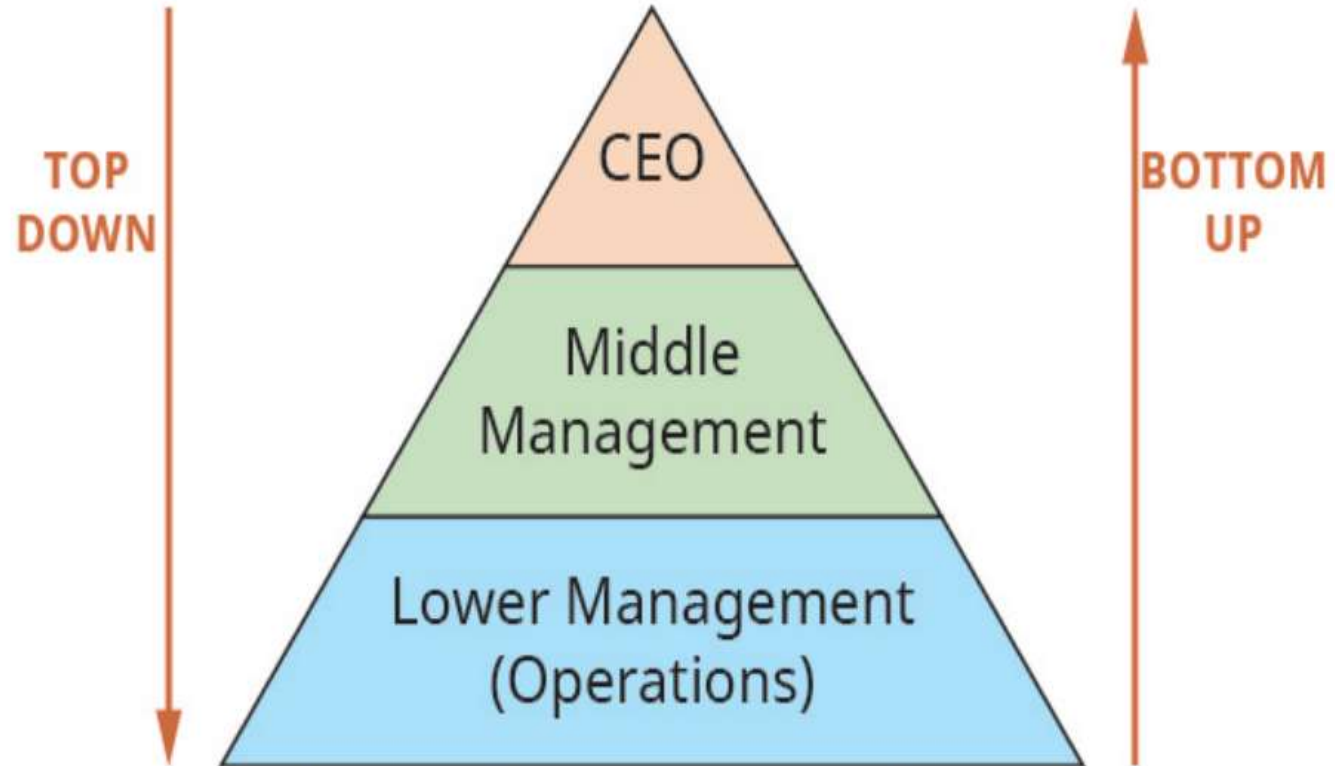
# Key concepts

## How are budgets developed?

Budgets can be derived from a **top-down** approach or from a **bottom-up** approach.

The top-down approach typically begins with senior management. The goals, assumptions, and predicted revenue and expenses information are passed from the senior manager to middle managers, who further pass the information downward.

Each department must then determine how it can allocate its expenses efficiently while still meeting the company goals.



*Top-Down versus Bottom-Up Approach to Budgeting. The top-down approach to budgeting starts with upper-level management, while the bottom-up approach starts with input from lower-level management. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)*

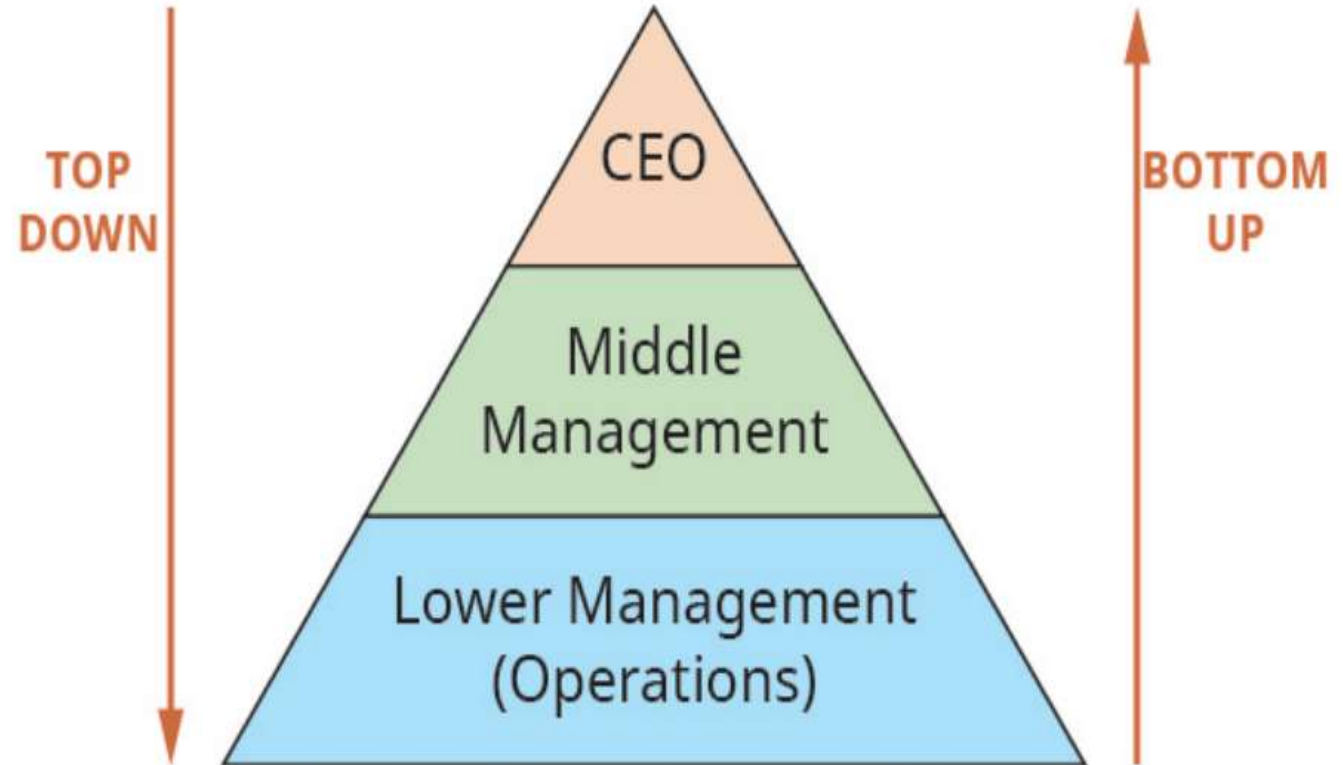
# Key concepts

## How are budgets developed?

The **benefit** of this approach is that it ties in to the strategic plan and company goals. Another benefit of passing the amount of allowed expenses downward is that the final anticipated costs are reduced by the vetting (fact checking and information gathering) process.

In the **top-down approach**, management must devote attention to efficiently allocating resources to ensure that expenses are not padded to create budgetary slack.

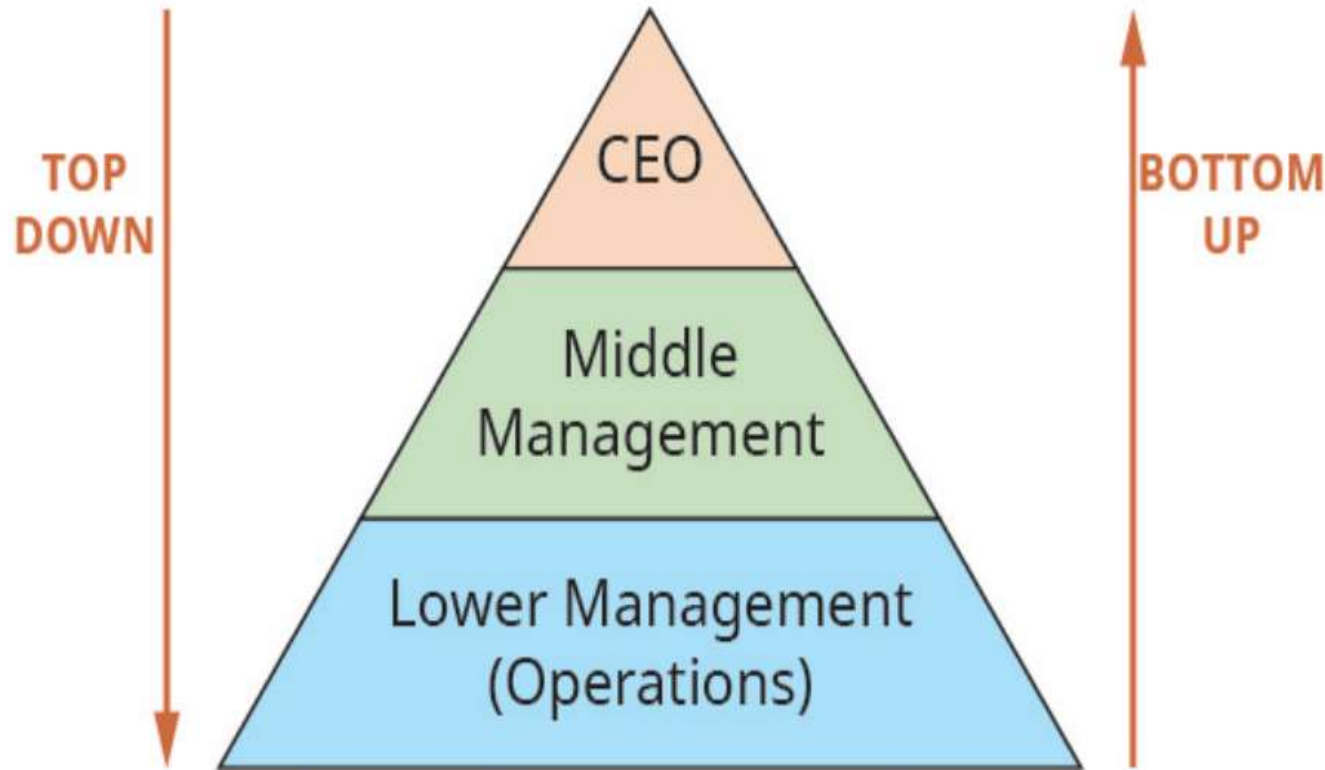
The **drawback** to this approach to budgeting is that the budget is prepared by individuals who are not familiar with specific operations and expenses to understand each department's nuances.



*Top-Down versus Bottom-Up Approach to Budgeting. The top-down approach to budgeting starts with upper-level management, while the bottom-up approach starts with input from lower-level management. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)*

# Key concepts

## How are budgets developed?



*Top-Down versus Bottom-Up Approach to Budgeting. The top-down approach to budgeting starts with upper-level management, while the bottom-up approach starts with input from lower-level management. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)*

The **bottom-up approach** (sometimes also named a self-imposed or participative budget) begins at the lowest level of the company. After senior management has communicated the expected departmental goals, the departments then plan and predict their sales and estimates the amount of resources needed to reach these goals. This information is communicated to the supervisor, who then passes it on to upper levels of management.

The **advantages** of this approach are that managers feel their work is valued and that knowledgeable individuals develop the budget with realistic numbers. Therefore, the budget is more likely to be attainable.

The **drawback** is that managers may not fully understand or may misunderstand the strategic plan.

# Key concepts

- Other approaches in addition to the top-down and bottom-up approaches include:
  - ✓ the **combination** approach – guidelines and targets are set at the top while the managers work to develop a budget within the targeted parameters.
  - ✓ **zero-based** budgeting approach – budgeting begins with zero dollars and then adds to the budget only revenues and expenses that can be supported or justified.
  - ✓ **rolling budget** approach – budget is adjusted monthly, and a new month is added as each month passes.

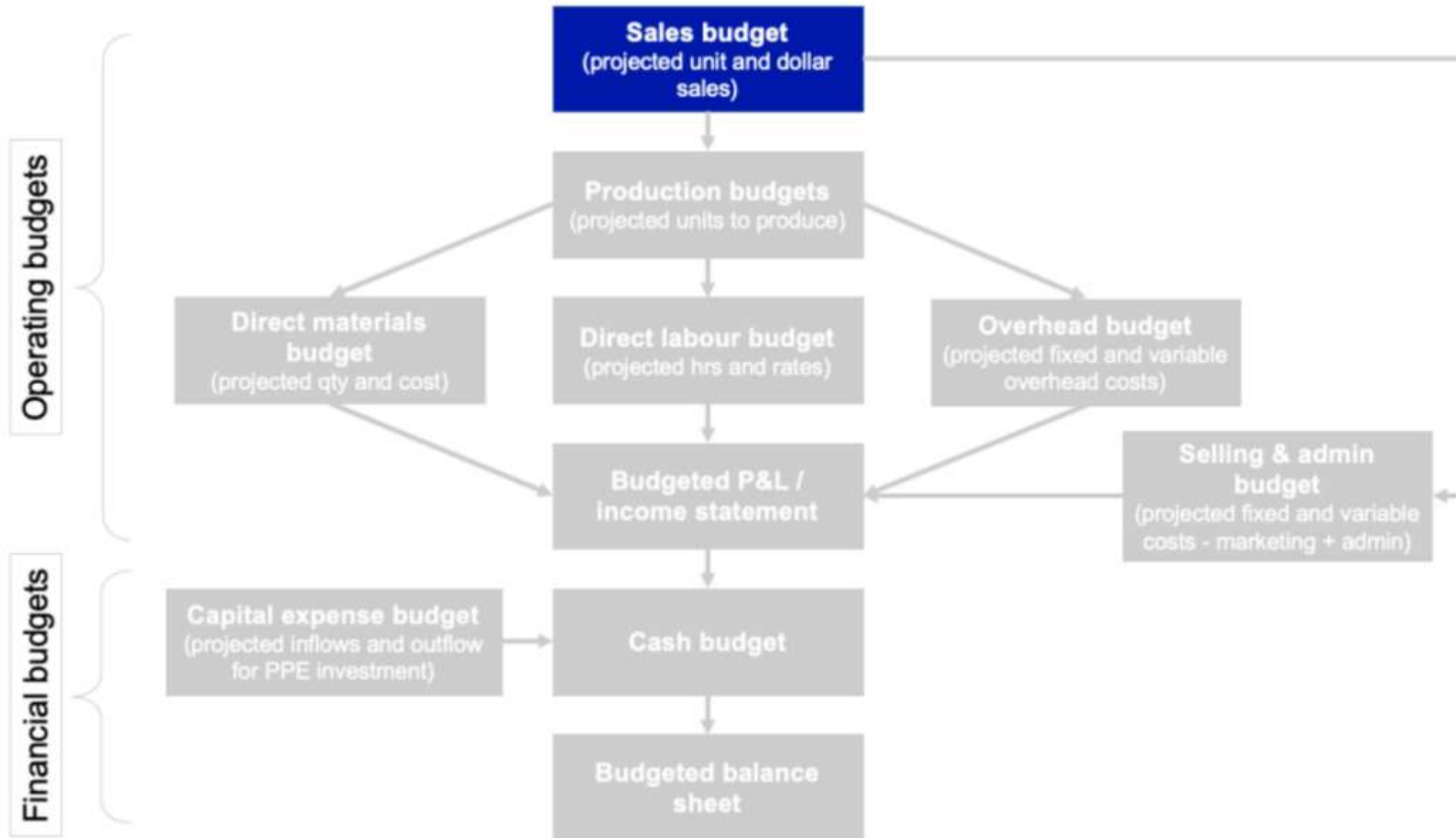
## Quick Check

**Why is the sales budget the first budget prepared in the master budget?**

- The sales budget is the most important budget.
- I don't know.
- The marketing team are the most enthusiastic team and usually have their plans ready first.
- The sales budget is used to create the rest of the budgets.

# Key concepts

## Preparing a sales budget



# Key concepts

## Preparing a sales budget

- The budgeting process begins with the estimate of sales. When management has a solid estimate of sales for each quarter, month, or other relevant time period, they can determine how many units must be produced. From there, they determine the expenditures, such as direct materials necessary to produce the units.
- The sales budget details the expected sales in units and the sales price for the budget period, it will use the following information to generate sales:
  - ✓ Sales activity for the business from previous years
  - ✓ Competitor sales activity
  - ✓ Industry trends
  - ✓ Economy-wide trends
  - ✓ Planned marketing campaigns
  - ✓ Weather

## Key concepts

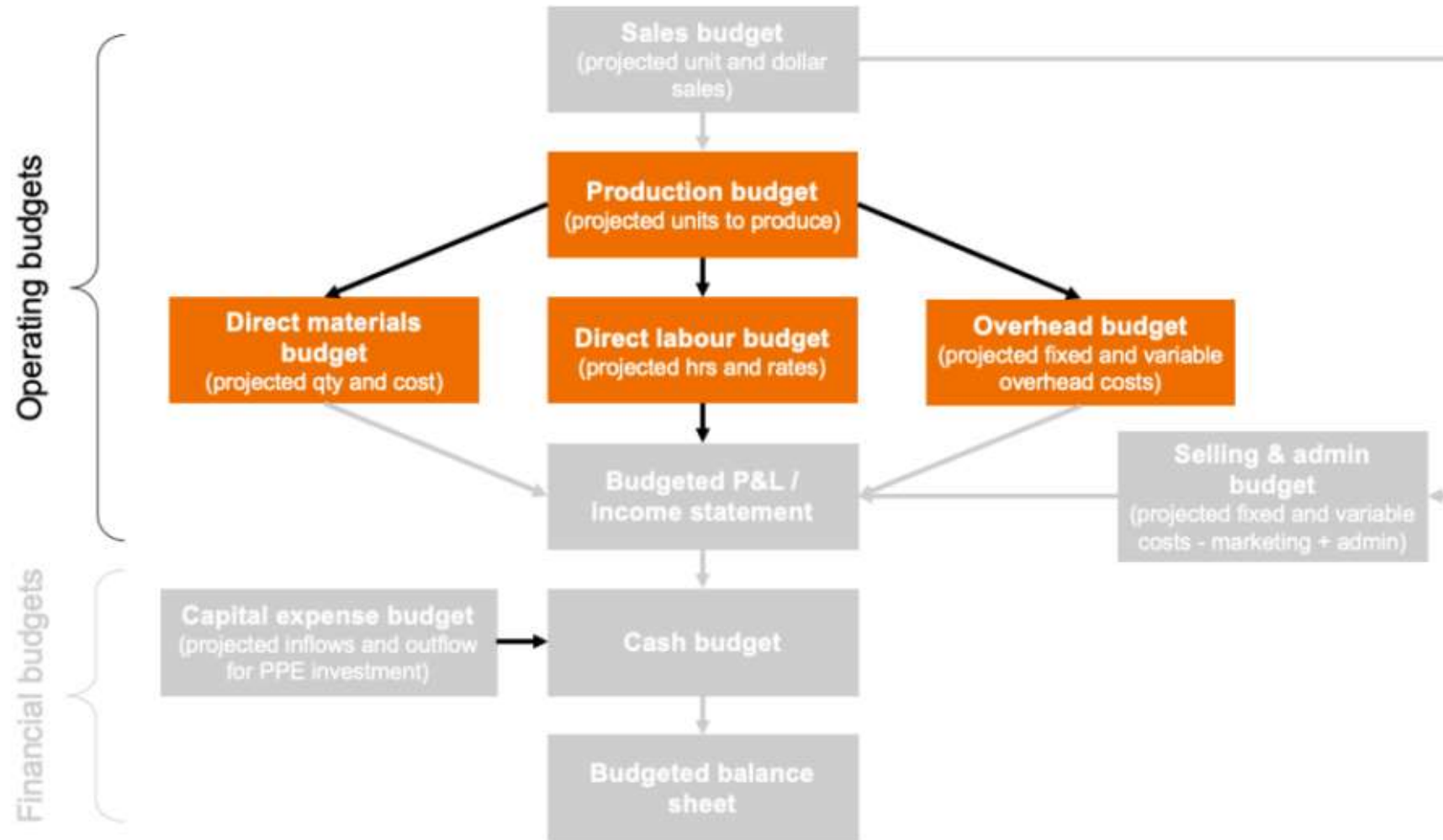
For example, Big Bad Bikes used the above information to estimate the number of units that will be sold in each quarter of the coming year. The number of units is multiplied by the sales price to determine the sales by quarter as shown in the table below.

| <b>BIG BAD BIKES</b><br><b>Sales Budget</b><br><b>For the Year Ended December 31, 2019</b> |                  |                  |                  |                  |                   |
|--------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|-------------------|
|                                                                                            | <b>Quarter 1</b> | <b>Quarter 2</b> | <b>Quarter 3</b> | <b>Quarter 4</b> | <b>Total</b>      |
| Expected Sales (Units)                                                                     | 1,000            | 1,000            | 1,500            | 2,500            | 6,000             |
| Sales Price per Unit                                                                       | \$ <u>70</u>     | \$ <u>70</u>     | \$ <u>75</u>     | \$ <u>75</u>     | <u>          </u> |
| Total Sales Revenue                                                                        | \$70,000         | \$70,000         | \$112,500        | \$187,500        | \$440,000         |

The sales budget leads into the production budget to determine how many units must be produced each week, month, quarter, or year. It also leads into the cash receipts budget.

# Key concepts

## Preparing production budgets



Operating Budgets, Financial Budgets, and the Relationship between Budgets. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license) / adjusted for Australian terminology and emphasis added

# Key concepts

## Preparing production budgets

- To be able to meet the demand predicted by the marketing department and management, the production or manufacturing department must be able to plan out how much inventory must be made and what components (materials, labour and overhead) will be required. A production budget plans out the inventory, and then sub-budgets related to materials, labour and overhead are produced.

# Key concepts

## Preparing production budgets

- Management wants to have enough inventory to meet production, but they do not want too much in the ending inventory to avoid paying for unnecessary storage. Management often uses a formula to estimate how much should remain in ending inventory.
- Management wants to be flexible with its budgeting, wants to create budgets that can grow or shrink as needed, and needs to have inventory on hand. So the amount of ending inventory often is a percentage of the next week's, month's, or quarter's sales.

### Cost of Goods Sold

Beginning Inventory  
+ Purchases (or produced)  
Goods available for sale  
– Ending Inventory

### Number of Units Produced

Goods Sold  
+ Ending Inventory  
Goods available for sale  
– Beginning Inventory

# Key concepts

## Preparing production budgets

- To illustrate the steps in developing a production budget, recall that Big Bad Bikes is introducing a new product that the marketing department thinks will have strong sales.
- For new products, Big Bad Bikes requires a target ending inventory of 30% of the next quarter's sales. Unfortunately, they were unable to manufacture any units before the end of the current year, so the first quarter's beginning inventory is 0 units.

| <b>BIG BAD BIKES</b><br><b>Production Budget</b><br><b>For the Year Ended December 31, 2019</b> |                  |                  |                  |                  |
|-------------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|
|                                                                                                 | <b>Quarter 1</b> | <b>Quarter 2</b> | <b>Quarter 3</b> | <b>Quarter 4</b> |
| Expected Sales                                                                                  | 1,000            | 1,000            | 1,500            | 2,500            |
| Desired Ending Inventory                                                                        | 300              | 450              | 750              | 1,050            |
| Total Required Units                                                                            | 1,300            | 1,450            | 2,250            | 3,550            |
| – Beginning Inventory                                                                           | 0                | 300              | 450              | 750              |
| Required Production                                                                             | 1,300            | 1,150            | 1,800            | 2,800            |

# Key concepts

## Preparing production budgets

The ending inventory from one quarter is the beginning inventory for the next quarter and the calculations are all the same. In order to determine the ending inventory in quarter 4, Big Bad Bikes must estimate the sales for the first quarter of the next year.

Big Bad Bikes's marketing department believes sales will increase in each of the next several quarters, and they estimate **sales as 3,500 for the first quarter of the next year and 4,500 for the second quarter of the next year**. Thirty percent of 3,500 is 1,050, so the number of units required in the ending inventory for quarter 4 is 1,050.

The number of units needed in production for the first quarter of the next year provides information needed for other budgets such as the direct materials budget, so Big Bad Bikes must also determine the number of units needed in production for that first quarter.

The estimated sales of 3,500 and the desired ending inventory of 1,350 (30% of the next quarter's estimated sales of 4,500) determines that 4,850 units are required during the quarter. The beginning inventory is estimated to be 1,050, which means the number of units that need to be produced during **the first quarter of year 2 is 3,800**.

# Key concepts

## Preparing a direct materials budget

From the production budget, management knows how many units need to be produced in each budget period. Management is already aware of how much material it needs to produce each unit and can combine the direct material per unit with the production budget to compute the direct materials budget.

This information is used to ensure the correct quantity of materials is ordered and the correct amount is budgeted for those materials.

Similar to the production budget, management wants to have an ending inventory available to ensure there are enough materials on hand. The direct materials budget illustrates how much material needs to be ordered and how much that material costs. The calculation is similar to that used in the production budget, with the addition of the cost per unit.

# Key concepts

## Preparing a direct materials budget

If Big Bad Bikes uses **3.2 pounds of material for each trainer** it manufactures and **each pound of material costs \$1.25**, we can create a direct materials budget. Management's goal is to **have 20% of the next quarter's material needs on hand** as the desired ending materials inventory. Recall that the **number of units to be produced during the first quarter of year 2 is 3,800**.

| BIG BAD BIKES<br>Direct Materials Budget<br>For the Year Ended December 31, 2019 |           |           |           |           |          |
|----------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|----------|
|                                                                                  | Quarter 1 | Quarter 2 | Quarter 3 | Quarter 4 | Total    |
| Units to be Produced                                                             | 1,300     | 1,150     | 1,800     | 2,800     | 7,050    |
| Direct Material per Unit                                                         | 3.20      | 3.20      | 3.20      | 3.20      | 3.20     |
| Total Pounds Needed for Production                                               | 4,160     | 3,680     | 5,760     | 8,960     | 22,560   |
| + Desired Ending Inventory                                                       | 736       | 1,152     | 1,792     | 2,432     | 2,432    |
| Total Material Required                                                          | 4,896     | 4,832     | 7,552     | 11,392    | 24,992   |
| – Beginning Inventory                                                            | 0         | 736       | 1,152     | 1,792     | 0        |
| Pounds of Direct Material Required                                               | 4,896     | 4,096     | 6,400     | 9,600     | 24,992   |
| Cost per Pound                                                                   | \$ 1.25   | \$ 1.25   | \$ 1.25   | \$ 1.25   | \$ 1.25  |
| Total Cost of Direct Material Purchase                                           | \$6,120   | \$5,120   | \$8,000   | \$12,000  | \$31,240 |

With 6,000 units estimated for sale (1050 ending inventory), 3.2 pounds of material per unit, and \$1.25 per pound, the direct materials used represent \$24,000 of the cost of goods sold. The remaining \$7,240 is included in ending inventory as units completed and raw material.

# Key concepts

## Preparing a direct labour budget

Management uses the same information in the production budget to develop the direct labour budget. This information is used to ensure that the proper amount of staff is available for production and that there is money available to pay for the labour, including potential overtime. Typically, the number of hours is computed and then multiplied by an hourly rate, so the total direct labour cost is known. If Big Bad Bikes knows that they need **45 minutes or 0.75 hours of direct labour for each unit** produced, and the labour rate for this type of manufacturing is **\$20 per hour**.

| <b>BIG BAD BIKES</b><br><b>Direct Labor Budget</b><br><b>For the Year Ended December 31, 2019</b> |                  |                  |                  |                  |              |
|---------------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|--------------|
|                                                                                                   | <b>Quarter 1</b> | <b>Quarter 2</b> | <b>Quarter 3</b> | <b>Quarter 4</b> | <b>Total</b> |
| Units to be Produced                                                                              | 1,300            | 1,150            | 1,800            | 2,800            | 7,050        |
| Direct Labor Hours per Unit                                                                       | 00.75            | 00.75            | 00.75            | 00.75            | 00.75        |
| Total Required Direct Labor Hours                                                                 | 975              | 862.50           | 1,350            | 2,100            | 5,287.50     |
| Labor Cost per Hour                                                                               | \$ 20            | \$ 20            | \$ 20            | \$ 20            | \$ 20        |
| Total Direct Labor Cost                                                                           | \$19,500         | \$17,250         | \$27,000         | \$42,000         | \$ 105,750   |

With 0.75 hours of direct labour per unit and \$20 per direct labour hour, each unit will cost \$15 in direct labour. Of the 7,050 units produced, 6,000 units will be sold, so \$90,000 represents the labour portion of the cost of goods sold and will be shown on the income statement, while the remaining \$15,750 will be the labour portion of ending inventory and will be shown on the balance sheet.

# Key concepts

## Preparing the overhead budget

The manufacturing overhead budget includes the remainder of the production costs not covered by the direct materials and direct labour budgets. In the manufacturing overhead budgeting process, producers will typically allocate overhead costs depending upon their cost behavior production characteristics, which are generally classified as either variable or fixed.

Based on this allocation process, the variable component will be treated as occurring proportionately in relation to budgeted activity, while the fixed component will be treated as remaining constant.

For Big Bad Bikes to create their manufacturing overhead budget, they first determine that the appropriate driver for **assigning overhead costs to products is direct labour hours**. The overhead allocation rates for the variable overhead costs are: **indirect material of \$1.00 per hour, indirect labour of \$1.25 per hour, maintenance of \$0.25 per hour, and utilities of \$0.50 per hour**. The fixed overhead costs per quarter are: **supervisor salaries of \$15,000, fixed maintenance salaries of \$4,000, insurance of \$7,000, and depreciation expenses of \$3,000**.

# Key concepts

## Preparing the overhead budget

| BIG BAD BIKES<br>Manufacturing Overhead Budget<br>For the Year Ended December 31, 2019 |           |           |           |           |           |
|----------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|
|                                                                                        | Quarter 1 | Quarter 2 | Quarter 3 | Quarter 4 | Total     |
| Variable Costs                                                                         |           |           |           |           |           |
| Indirect Material                                                                      | \$ 975    | \$ 863    | \$ 1,350  | \$ 2,100  | \$ 5,288  |
| Indirect Labor                                                                         | 1,219     | 1,078     | 1,688     | 2,625     | 6,609     |
| Maintenance                                                                            | 244       | 216       | 338       | 525       | 1,322     |
| Utilities                                                                              | 488       | 431       | 675       | 1,050     | 2,644     |
| Total Variable Manufacturing Costs                                                     | \$ 2,926  | \$ 2,588  | \$ 4,051  | \$ 6,300  | \$ 15,863 |
| Fixed Costs                                                                            |           |           |           |           |           |
| Supervisory Salaries                                                                   | \$15,000  | \$15,000  | \$15,000  | \$15,000  | \$ 60,000 |
| Maintenance Salaries                                                                   | 4,000     | 4,000     | 4,000     | 4,000     | 16,000    |
| Insurance                                                                              | 7,000     | 7,000     | 7,000     | 7,000     | 28,000    |
| Depreciation                                                                           | 3,000     | 3,000     | 3,000     | 3,000     | 12,000    |
| Total Fixed Manufacturing Costs                                                        | \$29,000  | \$29,000  | \$29,000  | \$29,000  | \$116,000 |
| Total Manufacturing Overhead                                                           | \$31,925  | \$31,588  | \$33,050  | \$35,300  | \$131,863 |

The total manufacturing overhead cost was \$131,863 for 7,050 units, or **\$18.70 per unit** (rounded).

Since 6,000 units are sold, \$112,200 (6,000 units × \$18.70 /unit) will be expensed as cost of goods sold, while the remaining \$19,663 will be part of finished goods ending inventory.

# Key concepts

## Preparing the selling and administration budget

The direct materials budget, the direct labour budget, and the manufacturing overhead budget plan for all costs related to production, while the selling and administrative expense budget contains a listing of variable and fixed expenses estimated to be incurred in all areas other than production costs.

If Big Bad Bikes pays a **sales commission of \$2 per unit** sold and a **transportation cost of \$0.50 per unit**, they can use these costs to put together their sales and administrative budget. All other costs are fixed costs per quarter: sales salaries of \$5,000; administrative salaries of \$5,000; marketing expenses of \$5,000; insurance of \$1,000; and depreciation of \$2,000.

# Key concepts

## Preparing the selling and administration budget

| <b>BIG BAD BIKES</b><br><b>Sales and Administrative Expense Budget</b><br><b>For the Year Ended December 31, 2019</b> |                  |                  |                  |                  |                 |
|-----------------------------------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|-----------------|
|                                                                                                                       | <b>Quarter 1</b> | <b>Quarter 2</b> | <b>Quarter 3</b> | <b>Quarter 4</b> | <b>Total</b>    |
| Budgeted Sales in Units                                                                                               | 1,000            | 1,000            | 1,500            | 2,500            | 6,000           |
| Variable Expenses                                                                                                     |                  |                  |                  |                  |                 |
| Sales Commissions                                                                                                     | 2,000            | 2,000            | 3,000            | 5,000            | 12,000          |
| Transportation                                                                                                        | 500              | 500              | 750              | 1,250            | 3,000           |
| Total Variable Expenses                                                                                               | <u>\$ 2,500</u>  | <u>\$ 2,500</u>  | <u>\$ 3,750</u>  | <u>\$ 6,250</u>  | <u>\$15,000</u> |
| Fixed Expenses                                                                                                        |                  |                  |                  |                  |                 |
| Sales Salaries                                                                                                        | \$ 5,000         | \$ 5,000         | \$ 5,000         | \$ 5,000         | \$20,000        |
| Administrative Salaries                                                                                               | 5,000            | 5,000            | 5,000            | 5,000            | 20,000          |
| Marketing Expenses                                                                                                    | 5,000            | 5,000            | 5,000            | 5,000            | 20,000          |
| Insurance Expenses                                                                                                    | 1,000            | 1,000            | 1,000            | 1,000            | 4,000           |
| Depreciation Expenses                                                                                                 | 2,000            | 2,000            | 2,000            | 2,000            | 8,000           |
| Total Fixed Expenses                                                                                                  | <u>\$18,000</u>  | <u>\$18,000</u>  | <u>\$18,000</u>  | <u>\$18,000</u>  | <u>\$72,000</u> |
| Total Selling and Administrative expenses                                                                             | <u>\$20,500</u>  | <u>\$20,500</u>  | <u>\$21,750</u>  | <u>\$24,250</u>  | <u>\$87,000</u> |

Only manufacturing costs are treated as a product cost and included in ending inventory, so all of the expenses in the sales and administrative budget are period expenses and included in the budgeted income statement.

# Key concepts

## Budgeted P&L/income statement

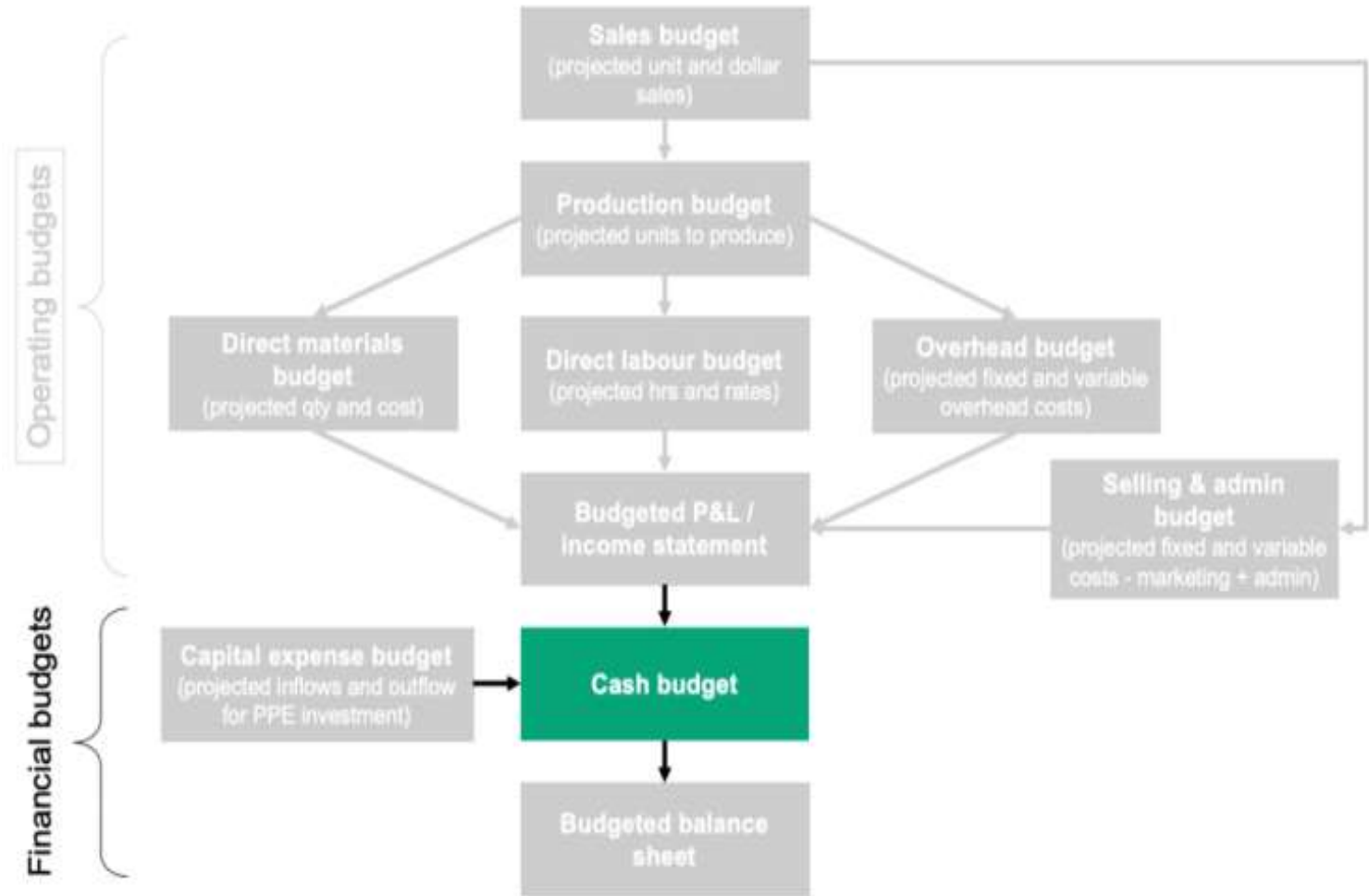
- A budgeted income statement is formatted similarly to a traditional income statement except that it contains budgeted data.

# Key concepts

## Preparing a cash budget

The financial budgets involves examining the expectations for financing the operations of the business and planning for the cash needs of the business. The budget helps estimate the source, amount, and timing of cash collection and cash payments as well as determine if and when additional financing is needed or debt can be paid.

We will focus on preparing the cash budget as shown in the master budget diagram below (preparing a capital expense budget and budgeted balance sheet is beyond the introductory scope of this text)

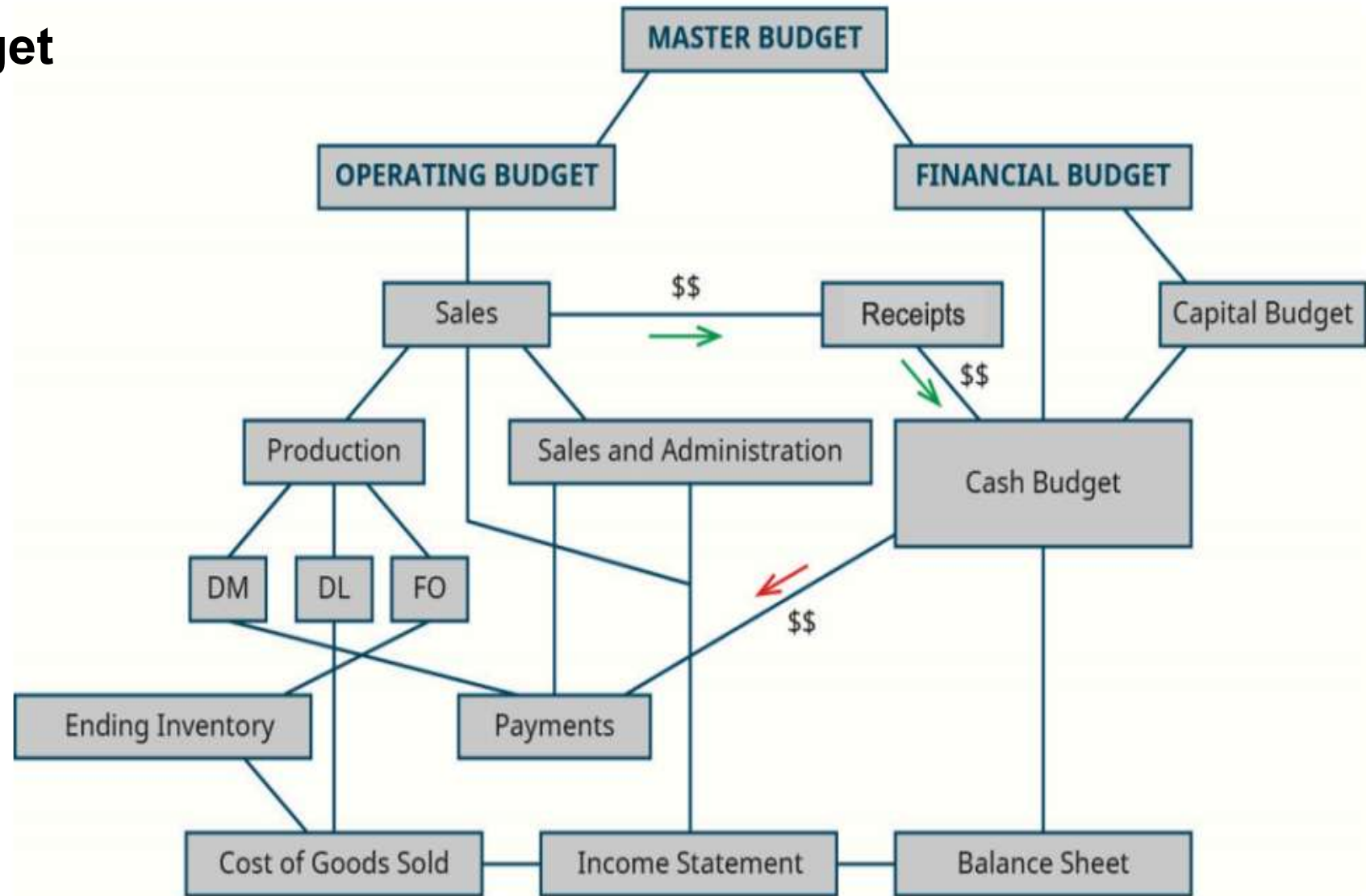


# Key concepts

## Preparing a cash budget

One goal of this budget is to anticipate the timing of cash inflows and outflows, which allows a company to try to avoid a decrease in the cash balance due to paying out more cash than it receives.

In order to provide timely feedback and alert management to short-term cash needs, the cash flow budget is commonly geared toward monthly or quarterly figures.



*Relationship between Budgets. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license) / amended for Australian terminology*

# Key concepts

## cash receipts budget

Because the cash budget accounts for every inflow and outflow of cash, it is broken down into smaller components.

The **cash receipts schedule** includes all of the cash inflow expected to be received from customer sales, whether those customers pay at the same rate or even if they pay at all. The cash receipts schedule includes all the cash expected to be received and does not include the amount of the receivables estimated as uncollectible.

In the quarter of sales, **65% of that quarter's sales will be collected**. In the quarter after the sale, **30% will be collected**. This **leaves 5% of the sales considered uncollectible**.

Uncollectible sales are those accounts receivables that cannot be collected and must be converted from an asset (Accounts receivable) to an Expense. These uncollectible sales are called bad debts and we will explore these in more detail in the follow up textbook, Accounting Business and Society.

# Key concepts

## cash receipts budget

| Percentage of Sales Collected |           |           |           |           |
|-------------------------------|-----------|-----------|-----------|-----------|
|                               | Quarter 1 | Quarter 2 | Quarter 3 | Quarter 4 |
| Prior year, Quarter 4 Sales   | 30%       |           |           |           |
| Quarter 1 Sales               | 65%       | 30%       |           |           |
| Quarter 2 Sales               |           | 65%       | 30%       |           |
| Quarter 3 Sales               |           |           | 65%       | 30%       |
| Quarter 4 Sales               |           |           |           | 65%       |

| BIG BAD BIKES<br>Cash Collections Schedule<br>For the Year Ended December 31, 2019 |           |           |           |           |           |           |
|------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|
|                                                                                    | Sales     | Quarter 1 | Quarter 2 | Quarter 3 | Quarter 4 | Total     |
| Collections from prior year Quarter 4                                              | 0         | 0         |           |           |           |           |
| Quarter 1                                                                          | \$ 70,000 | \$45,500  | \$21,000  |           |           | \$ 66,500 |
| Quarter 2                                                                          | 70,000    |           | 45,500    | \$21,000  |           | 66,500    |
| Quarter 3                                                                          | 112,500   |           |           | 73,125    | \$ 33,750 | 106,875   |
| Quarter 4                                                                          | 187,500   |           |           |           | 121,875   | 121,875   |
| Total Collections                                                                  | \$440,000 | \$45,500  | \$66,500  | \$94,125  | \$155,625 | \$361,750 |
| Accounts Receivable                                                                | \$ 78,250 |           |           |           |           |           |

# Key concepts

## cash payment budget

The cash payments schedule shows when cash will be used to pay for Accounts Payable. One such example are direct material purchases, which originates from the direct materials budget.

Big Bad Bikes typically **pays half of its purchases in the quarter of purchase**. The **remaining half is paid in the following quarter**, so payments in the first quarter include payments for purchases made during the first quarter as well as half of the purchases for the preceding quarter.

# Key concepts

## cash payment budget

|                                 | Percentage of Cash Payments for Purchases |           |           |           |
|---------------------------------|-------------------------------------------|-----------|-----------|-----------|
|                                 | Quarter 1                                 | Quarter 2 | Quarter 3 | Quarter 4 |
| Prior year, Quarter 4 Purchases | 50%                                       |           |           |           |
| Quarter 1 Purchases             | 50%                                       | 50%       |           |           |
| Quarter 2 Purchases             |                                           | 50%       | 50%       |           |
| Quarter 3 Purchases             |                                           |           | 50%       | 50%       |
| Quarter 4 Purchases             |                                           |           |           | 50%       |

| <b>BIG BAD BIKES</b><br><b>Cash Payments Schedule</b><br><b>For the Year Ended December 31, 2019</b> |           |           |           |           |           |          |
|------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|----------|
| Payments                                                                                             | Purchases | Quarter 1 | Quarter 2 | Quarter 3 | Quarter 4 | Total    |
| Quarter 4, prior year                                                                                | \$ 0      | \$ 0      |           |           |           |          |
| Quarter 1                                                                                            | 6,120     | 3,060     | \$3,060   |           |           | \$ 6,120 |
| Quarter 2                                                                                            | 5,120     |           | 2,560     | \$2,560   |           | 5,120    |
| Quarter 3                                                                                            | 8,000     |           |           | 4,000     | \$ 4,000  | 8,000    |
| Quarter 4                                                                                            | 12,000    |           |           |           | 6,000     | 6,000    |
| Total payments                                                                                       | \$31,240  | \$3,060   | \$5,620   | \$6,560   | \$10,000  | \$25,240 |
| Accounts Payable                                                                                     | \$ 6,000  |           |           |           |           |          |

# Key concepts

While the cash payments schedule is made for purchases of material on account, there are other outflows of cash for the company, and management must estimate all other cash payments for the year. Typically, this includes the manufacturing overhead budget, the sales and administrative budget, the capital asset budget, and any other potential payments of cash. Since depreciation is an expense not requiring cash, the cash budget includes the amount from the budgets less depreciation. Cash payments are listed on the **cash budget** following cash receipts.

## JOB COST SHEET General Overview of Cash Budget Components\*

Cash Receipts from Sales  
+ Other cash receipts (issuance of stock, borrowing money, receiving interest or dividends, from selling assets such as equipment, etc.)  
- Cash Payments for Purchases or Production of Inventory  
- Cash Payments for manufacturing expenses\*\*  
- Cash Payments for selling and administrative expenses\*\*  
- Cash payments for capital asset purchases  
- Other cash payments (paying interest, paying loan payments, etc.)  
= Net Cash

\*This is a general overview of the types of cash transactions that might appear in a cash budget and is representative of the components but not of a typical presentation of those components

\*\*Note that depreciation, a non-cash expense, would be excluded from these expenses

# Key concepts

## cash budget

| <b>BIG BAD BIKES</b><br><b>Cash Budget</b><br><b>For the Year Ended December 31, 2019</b>             |            |           |           |           |           |
|-------------------------------------------------------------------------------------------------------|------------|-----------|-----------|-----------|-----------|
|                                                                                                       | Quarter 1  | Quarter 2 | Quarter 3 | Quarter 4 | Total     |
| Beginning Cash Balance                                                                                | \$13,000   | \$10,000  | \$ 10,000 | \$ 10,000 | \$ 13,000 |
| Collections from Customers<br>(Cash Collections Schedule)                                             | 45,500     | 66,500    | 94,125    | 155,625   | 361,750   |
| Issuing of Stock                                                                                      | 5,000      |           |           |           | 5,000     |
| Total Cash Collected during the Period                                                                | \$50,500   | \$66,500  | \$ 94,125 | \$155,625 | \$366,750 |
| Total Available Cash                                                                                  | \$63,500   | \$76,500  | \$104,125 | \$165,625 | \$379,750 |
| – Disbursements                                                                                       |            |           |           |           |           |
| Direct Materials (Cash Payments Schedule)                                                             | 3,060      | 5,620     | 6,560     | 10,000    | 25,240    |
| Direct Labor (Direct Labor Budget)                                                                    | 19,500     | 17,250    | 27,000    | 42,000    | 105,750   |
| Manufacturing Overhead Less Depreciation<br>(MFG OH Budget)                                           | 28,925     | 28,588    | 30,050    | 32,300    | 119,863   |
| Selling and Administrative Expenses Less<br>Depreciation (Sales and Administrative<br>Expense Budget) | 18,500     | 18,500    | 19,750    | 22,250    | 79,000    |
| Income Tax Expense                                                                                    | 1,000      | 1,000     | 1,000     | 1,000     | 4,000     |
| Purchase of Copier (Capital Asset Budget)                                                             |            |           | 8,500     |           | 8,500     |
| Total Disbursements                                                                                   | \$70,985   | \$70,958  | \$ 92,860 | \$107,550 | \$342,353 |
| Excess (deficiency) of Available Cash                                                                 | (\$ 7,485) | \$ 5,542  | \$ 11,265 | \$ 58,075 | \$ 37,397 |
| Financing                                                                                             |            |           |           |           |           |
| + Borrowings                                                                                          | 17,485     | 4,458     |           |           | 21,943    |
| – Repayments Including Interest                                                                       |            |           | (1,265)   | (21,632)  | (22,897)  |
| Ending Cash Balance                                                                                   | \$10,000   | \$10,000  | \$ 10,000 | \$ 36,443 | \$ 36,443 |

Organizations typically require a **minimum cash balance**. If the potential ending cash balance does not meet the minimum amount, management must plan to acquire financing to reach that amount. If the potential ending cash balance exceeds the minimum cash balance, the excess amount may be used to pay any financing loans and interest.

Big Bad Bikes has a **minimum cash balance** requirement of **\$10,000** and the estimated interest payment for the year is **\$954**. They also plan to issue additional **capital stock for \$5,000 in the first quarter**, to pay **taxes of \$1,000** during each quarter, and to **purchase a copier for \$8,500 cash in the third quarter**. The **beginning cash balance** for Big Bad Bikes is **\$13,000**, which can be used to create the cash budget.

## Quick Check

**Which of the answers below is not included on the cash budget?**

- Cash payments
- Cash receipts
- Cost of sales expense
- Opening cash balance

# Key concepts

## **Budgeted balance sheet**

- The cash budget shows how cash changes from the beginning of the year to the end of the year, and the ending cash balance is the amount shown on the budgeted balance sheet.
- The budgeted balance sheet is the estimated assets, liabilities, and equities that the company would have at the end of the year if their performance were to meet its expectations.
- Creating a budgeted balance sheet is a more advanced skill not covered in this text.

# Key concepts

## Using budgets to evaluate performance

- Comparing budget vs actual performance, is it a fair deal?
- Our budgets are created using a number of assumptions including those related to:
  - ✓ customer demand
  - ✓ economic sentiment
  - ✓ price of raw materials
  - ✓ availability of raw materials
  - ✓ pay rates of labour
  - ✓ prices of overheads and selling and administration costs

# Key concepts

## Using budgets to evaluate performance

- However, as the COVID19 pandemic clearly illustrated, budgets and plans may need to be adjusted due to changes in an industry, a country's economic situation or in reaction to global events.
- The original budget produced for planning is called a **static budget**.
- The businesses use a technique called **flexible budgeting** – where an original budget can be adjusted for changes in sales volume or activity.
- A flexible budget adjusts the cost of goods produced for varying levels of production and is more useful than a static budget, which remains at one amount regardless of the production level.
- A flexible budget is created at the end of the accounting period, whereas the static budget is created before the fiscal year begins.

## Tutorial activities--Tutorial 9A

A business plans to sell 2,000 units in quarter 1 and 3,000 units in quarter 2. The selling price for quarter 1 is \$100. They plan to increase the price by 10% in quarter 2. What is the sales budget for quarter 2?

- a. 3,000
- b. \$200,000
- c. \$300,000
- d. \$330,000

A business plans to sell 2,000 units in quarter 1 and 3,000 units in quarter 2. They like to keep 10% of sales for next quarter in ending inventory. What is the production budget for quarter 1?

- a. 2,100
- b. 2,300
- c. 1,900
- d. There is not enough information to complete this question.

## Tutorial activities--Tutorial 9A

A business has a production budget of 2,100 in quarter 1 and 2,800 in quarter 2. Each product needs five parts, which cost \$2 each. They like to keep 15% of the materials they need next quarter in their ending inventory. How many parts do they need to buy and how much will this cost in quarter 1?

- a. 12,600 parts, \$25,200
- b. 9,975 parts, \$19,950
- c. 2,205 parts, \$4,410
- d. 11,025 parts, \$22,050

## Tutorial activities--Tutorial 9A

A business starts the quarter with \$6,800 in cash. They make sales to customers of \$20,000 and collect \$9,500 in cash. They also collect \$8,000 from accounts receivable. The business buys \$11,000 of materials but only pays for 70% of the purchases as well as \$3,000 from last month's purchases. The business pays for labour \$4,000, overhead \$2,000 and selling and administration costs of \$1,000. The business would like to keep a minimum cash balance of \$10,000. If the closing balance is less than \$10,000 then the business will borrow enough money from the bank to keep the minimum balance. The business can only borrow in \$500 amounts.

How much cash must be borrowed at the end of the quarter to ensure the business has the minimum cash balance of \$10,000?

- a.\$6,600
- b.\$3,400
- c.\$3,500
- d.\$3,000

# Tutorial activities--Tutorial 9A Budgeting

Marketing Love ❤️ makes and sells corporate gift baskets that clients can use for marketing campaigns and gifts.

|              | Q 1 (July – Sept) | Q 2 (Oct – Dec) | Q 3 (Jan – Mar) | Q 4 (April – June) |
|--------------|-------------------|-----------------|-----------------|--------------------|
| Budget sales | 1,200             | 2,200           | 800             | 1,000              |

1. Sales budget: The baskets sell for \$100 each.
2. Production budget: The business has 120 baskets in beginning inventory in quarter 1. Management wants ending inventory at the end of the quarter to be 10% of next quarters predicted sales. Sales for quarter 1 and 2 of the next year are expected to be 1,400 and 3,000 respectively.
3. Direct materials budget: To make the baskets they need 6 items that cost on average \$5 each. Management wants the ending inventory of items to be 5% of what is needed the next quarter. The beginning inventory at the start of quarter 1 is 390 items.
4. Direct labour budget: It takes 12 minutes to make one basket and the labour cost is \$30 per hour.
5. Overhead budget: The variable overhead is \$6 per basket made. This includes electricity used and supplies used. The fixed overhead per quarter includes \$12,000 for rent and \$9,000 for the supervisors salary.
6. Selling and administration budget: The selling and administration costs includes a sales commission of \$1 per pack sold and quarterly advertising costs of \$2,400 and website costs of \$1,000.
7. Cash budget: The opening cash balance is \$11,500 in quarter 1. All sales are paid for using direct bank deposit before the baskets are sent out (this means there is no accounts receivable to collect). When the basket items are purchased, they pay 30% in the current quarter and the rest in the next quarter. Labour and overheads are paid in the quarter they are incurred. The selling and administration costs are paid in the quarter after they are incurred. At the start of quarter 1 they owe \$16,800 to basket item suppliers and \$4,400 in selling and administration costs.

# Tutorial activities--Tutorial 9A Budgeting

Marketing Love ❤️ makes and sells corporate gift baskets that clients can use for marketing campaigns and gifts.

|              | Q 1 (July – Sept) | Q 2 (Oct – Dec) | Q 3 (Jan – Mar) | Q 4 (April – June) |
|--------------|-------------------|-----------------|-----------------|--------------------|
| Budget sales | 1,200             | 2,200           | 800             | 1,000              |

## Required:

Using this business information prepare all the budgets for quarter 1 – 4 this year.

Then using the cash budget answer these questions:

- 1.The business is considering spending \$100,000 to expand and grow. When would you recommend they spend this money?
- 2.The business likes to keep a closing balance of \$70,000 in cash each quarter and can borrow in lots of \$1,000. Will they need to borrow any money?

## Tutorial activities--Tutorial 9B

Lola makes and sells beaded necklaces. Here is the sales budget:

|              | Q 1 (July – Sept) | Q 2 (Oct – Dec) | Q 3 (Jan – Mar) | Q 4 (April – June) |
|--------------|-------------------|-----------------|-----------------|--------------------|
| Budget sales | 120               | 200             | 150             | 90                 |

### Question 1

Production budget: Lola wants ending inventory at the end of the quarter to be 10% of next quarters predicted sales. Sales for quarter 1 and 2 of the next year are expected to be 130 and 220 respectively.

Calculate how many units Lola needs to make in quarter 2?

## Tutorial activities--Tutorial 9B

Lola has the following production budget:

|                   | Q 1 (July – Sept) | Q 2 (Oct – Dec) | Q 3 (Jan – Mar) | Q 4 (April – Jun) |
|-------------------|-------------------|-----------------|-----------------|-------------------|
| Production budget | 128               | 195             | 144             | 94                |

### Question 2

Direct materials budget: To make each necklace she needs 10 beads. Each bead costs on average \$3 each. Due to supply chain issues, Lola wants the ending inventory to be 20% of the items needed the next quarter, to ensure she doesn't run out.

How many beads does Lola need to purchase in quarter 2 and how much will this cost?

## Tutorial activities--Tutorial 9B

Lola has the following production budget:

|                   | Q 1 (July – Sept) | Q 2 (Oct – Dec) | Q 3 (Jan – Mar) | Q 4 (April – Jun) |
|-------------------|-------------------|-----------------|-----------------|-------------------|
| Production budget | 128               | 195             | 144             | 94                |

### Question 3

Direct labour budget: Each necklace takes one hour to make and labour costs \$20 per hour. How many hours of labour are required in quarter 4 and how much will this cost?

# Tutorial activities--Tutorial 9B

## Question 4

**Calculate the closing cash balance for quarter 1 using the information provided below:**

- The opening cash balance is \$15,000 in quarter 1.
- All sales are made on credit. 80% of the sales are collected in the quarter they were made and 20% are collected in the next quarter. Sales in quarter one were \$7,200.
- When the parts are purchased, we pay 30% in the current quarter and the rest in the next quarter. The cost of direct materials in quarter one were \$4,242.
- Labour and overheads are paid in the quarter they are incurred. The costs in quarter one were \$2,560 for labour and \$3,556 for overhead.
- The selling and administration costs for quarter one are \$3,100. These are paid in the quarter after they are incurred.
- At the start of quarter 1 they are waiting to collect \$2,000 from accounts receivable, owe \$1,600 to parts suppliers and \$6,000 in selling and administration costs.

# Tutorial problems week 9

Follow the guide of assumptions in the previous Lola's beaded necklaces business, design your own little business, prepare a master budget for the first year and present your findings.

- You will work in groups (4-5 in one group) on tutorial problems.
- You will prepare a PPT and present the solution with the class in tutorial week.
- All group members need to participate and explain the solution and also answer any further questions from the class.
- Time limit: 15-20 minutes.

# Assessment

**Assessment 2B Written Reflection** is due at the end of week 13.

Review what you wrote in Assessment 2A Written reflection, analyse your goals and what has happened and then plan for the rest of the semester.

In this assignment, you will write about what you have learned this semester and how this may be useful in your career. You will also reflect on your achievements and progress in studying this semester and prepare for your final exam.

# Assessment

## Assessment 2B Written Reflection

### 1. Career goals and accounting

Summarise your career goals from Assessment 2A and discuss whether you initially expected accounting to be relevant (in two short sentences). Explain how your career goals and views about accounting changed over the semester.

### 2. Learning impact

Explain something you learnt to do this semester and how it could be useful for your career.

### 3. SMART goals and performance evaluation

Compare your actual grades to your grade goal. What has worked and what hasn't worked for you as you try to achieve your grade goal? How do you feel about your progress so far?

### 4. Final exam preparation

Describe your feelings and emotions about the upcoming final exam. How do you plan to prepare? Did writing this reflection help you learn anything that might assist in your exam preparation?